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SEASONAL PRODUCTION DYNAMICS IN A GUILD OF PERIPHYTON-GRAZING INSECTS IN A SOUTHERN APPALACHIAN STREAM[1]

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Abstract. Temporal partitioning has been found to be a predominant mode of ecological segregation among groups of systematically related stream insects. We extended this concept to a functionally similar but systematically diverse group of species. The life cycles and secondary production of six species of periphyton-grazing insects (scrapers) were studied in a fourth-order unshaded stream reach in the southern Appalachian Mountains.

Total annual production (as ash-free dry mass) calculated with the instantaneous-growth rate method was 1195 mg/m². The size-frequency estimate was 6.4% lower. Annual production (AFDM) by species ranged from 612 mg/m² for *Glossosoma nigrior* to 10 mg/m² for *Goera fuscula*. The production peaks of the six species occurred at separate points in the year, with very little overlap between species. Comparison with a neutral model indicated that production peaks were significantly more regularly spaced ($P < .1$) than would be expected by chance alone. Temporal overlap between species was calculated using density, biomass, and production. Overlaps based on density and biomass were similar (means of .173 and .171, respectively). Mean overlap based on production was significantly lower (.124).

The patterns of temporal organization observed in this group of species were linked with specialized life histories. Five of the six species were univoltine, with periods of larval development (cohort production intervals [CPI]) ranging from 44 to 215 d. The shortest CPIs were associated with species that diapaused in the egg (*Agapetus* sp.) or larval (*Neophylax consimilis*) stages. Summer production was dominated by *G. nigrior*, which had a bivoltine life cycle. Production by the spring-summer cohort of *G. nigrior* was 4.5× the production of the winter cohort. *Glossosoma*'s bivoltinism apparently provides it with the life history flexibility to utilize the high summer periphyton production permitted by the lack of canopy cover at the study site.

Seasonal production by the six species was distributed 18% in winter (January–March), 36% in spring, 38% in summer, and 8% in autumn. The seasonal distribution of grazer secondary production correlates well with previously reported seasonal dynamics of periphyton production.

Key words: Diptera; life cycles; phenology; production; scrapers; stream invertebrates; temporal organization; Trichoptera.

INTRODUCTION

Our understanding of the structure and function of stream ecosystems has advanced rapidly since Cummins (1973) presented a conceptual model in which invertebrates were organized into functional feeding groups. Functional relationships provide a priori definitions of stream invertebrate guilds that are independent of the taxonomic affinities of the member species (Jaksić 1981). The river continuum concept (Vannote et al. 1980) identified physical and biological changes that occur along longitudinal gradients in streams. The continuum hypothesis postulates that the relative importance of functional groups will vary from headwater streams to large rivers in response to shifts in stream morphology, temperature regime, and energy input.

Minshall (1978) argued that autochthonous primary production is the major source of organic matter in many streams, and that even in streams characterized as heterotrophic, autochthonous production plays an important role. The importance of epilithic periphyton production may be obscured by very low biomass, which supports large grazer standing crops by rapid turnover rates (McIntire and Colby 1978). Consideration of the food quality of periphyton, as compared to leaf detritus, further elevates the potential trophic importance of autochthonous production (Winterbourn 1971, McCullough et al. 1979, Hornick et al. 1981).

The predictions of the continuum concept suggest that the structure of stream scraper guilds may best be studied in unshaded streams of intermediate size where autochthonous primary production peaks (Vannote et al. 1980). Hawkins and Sedell (1981) found that scraper abundance reached a maximum in fifth-order Lookout Creek (Oregon) and declined slightly down-

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stream in seventh-order McKenzie River. The longitudinal distribution of scrapers paralleled roughly the pattern of primary production, but the correlation between the two was not significant. In all seasons but spring, gross primary production was higher in McKenzie River than in Lookout Creek.

Vannote and Sweeney (1980) hypothesized that as a result of competition for food, the periods of maximum resource use by species in a given stream functional group will follow a seasonal sequence, and resource overlap among species within functional groups will be minimized. Periphyton productivity responds to both light and temperature (McIntire and Colby 1978). These factors normally reach maximum levels in the summer, but the presence of a closed canopy over smaller streams shifts the maxima to the spring and fall. If interactions among species within the scraper guild result in a close relationship between resource availability and use, scraper secondary production would be expected to peak in the spring and fall, with maximum production by each species occurring at a different period of the year.

We studied the life cycles and secondary production of six species of scrapers in a fourth-order unshaded stream reach in the southern Appalachian Mountains. Preliminary sampling (bimonthly, 1978–1979) provided a list of the scrapers present. Scrapers of epilithic periphyton were selected, based on trophic and functional categories (Merritt and Cummins 1978) and habitat preferences. The sampling methods we used were designed in connection with an analysis of microhabitat distributions and were efficient only in the collection of organisms confined to epilithic habitats. While it is known that some collector-gatherers, primarily mayflies, ingest significant quantities of periphyton, their high mobility precluded their inclusion in this study.

Scrapers from the Trichoptera, Diptera, and Coleoptera were present. The caddisflies were *Agapetus* sp., *Glossosoma nigrior* Banks (Glossosomatidae), *Goera fuscula* Banks, *Neophylax consimilis* Bett., and *N. mitchelli* Carpenter (Limnephilidae). The Diptera were all *Blepharicera* (Blephariceridae) species. *B. williamsae* Alexander larvae were identifiable (Hogue 1978, C. L. Hogue, *personal communication*). At least three other species were present, most of which were tentatively identified as *B. similans* (Johannsen) (Hogue 1978, C. L. Hogue, *personal communication*), but many individuals intermediate between *B. similans* and *B. cherokea* were found. In addition, two individuals of Hogue's (1978) very distinctive "type F" larvae were collected. Specimens not identifiable as *B. williamsae* were combined into a *Blepharicera* spp. category. *Neophylax mitchelli* and three coleopteran species were present in very low abundances and were not included in the study.

The selected species formed a homogeneous group in terms of feeding mode and habitat preferences. The objectives of this study are to: (1) describe the life cycles and seasonal secondary production of these species, (2) determine if seasonal patterns of abundance and production are important elements of community structure in this group of grazers, and (3) compare the role of the grazer functional group with other functional groups in similar mountainous streams in the eastern deciduous biome.

Methods

Study site

The United States Forest Service's Coweeta Hydrologic Laboratory (Macon County, North Carolina) occupies a 1625-ha drainage basin in the Blue Ridge Province of the southern Appalachian Mountains of western North Carolina. Basin vegetation consists of mixed-hardwood forest, dominated by *Quercus* and *Carya* species. Stream margins are lined with dense growths of rhododendron (*Rhododendron maximum* L.), mountain laurel (*Kalmia latifolia* L.), and eastern hemlock (*Tsuga canadensis* [L.] Carr.). The Coweeta basin has not been subjected to timber cutting or other nonexperimental disturbances since 1924 (Swank and Douglass 1975). A variety of experimental manipulations, including clear-cutting, selective logging, grazing, and herbicide treatments, have occurred since the 1930s (see Swank and Douglass [1975] for summaries). Most recently, a 59-ha basin ≈0.9 km upstream of the study site was commercially clear-cut in January–May 1977 (Gurtz 1981).

Sampling sites were established along a 200-m reach of Lower Shope Fork that flows through an open meadow. All riparian vegetation is cut periodically by the Forest Service along the unshaded reach. The riparian vegetation is composed of small growths of alder (*Alnus serrulata* [Aiton] Willd.), raspberry (*Rubus* spp.), and various annuals. The stream at the study site is ≈5.5 m wide (mean 5.49 m, range 3.20–7.45 m) and 0.15 m deep (mean 0.16, maximum depths of ≈1.5 m). Riffles predominate, with no distinct pools. The substrate is rocky, with boulders (>0.25 m diameter) common, and median substrate size (on an areal basis) is 10.5 cm.

The drainage basin at the sample site is ≈800 ha. Weirs on the main stream and on two small tributaries that enter above the sample site provide continuous measurement of discharge. Monthly average discharges (Fig. 1) during the summer of 1980, a drought year, were below long-term averages during the last two months of the sampling period. Water temperature, monitored biweekly with minimum/maximum thermometers at the upstream and downstream margins of the study site, is plotted in Fig. 1.

Lower Shope Fork has low ionic concentrations of biologically important elements: K, 0.386 mg/L; Ca, 0.804 mg/L; Mg, 0.323 mg/L; nitrate-N, 9 μg/L; and phosphate-P, 2 μg/L (Swank and Douglass 1975), and total alkalinity (as $CaCO_3$) of 10 mg/L.

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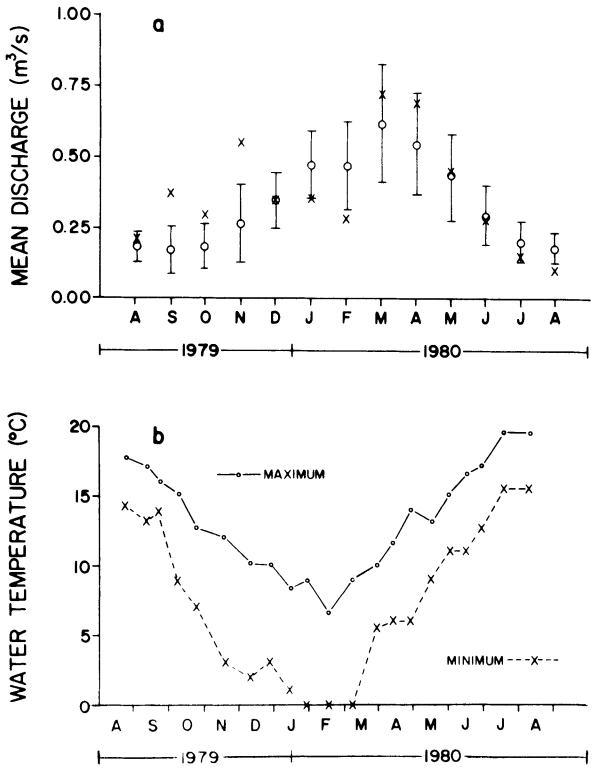

FIG. 1. Some physical characteristics of Lower Shope Fork during the study period. (a) Monthly mean discharge during study period (×) and 10-yr average discharge (monthly mean ± 1 SD indicated by open circles and vertical lines). (b) Water temperatures during the study period (maximum and minimum temperatures over ≈2-wk intervals between sampling dates).

Benthic samples

Densities and standing stocks were measured for the six most abundant species of grazers present at the study site. Fifteen benthic quadrat samples, each 0.10 m^2, were collected approximately biweekly from August 1979 to August 1980. In order to permit rapid separation of specimens from the samples in the field, a rock-surface sampling method was adopted with minimum disturbance of the underlying sand–fine-gravel substrate. Rock-surface collections are superior to other sampling methods for early-instar glossosomatid species, since their minute sand-grain cases are difficult to distinguish from a sandy-gravel substrate (Anderson and Bourne 1974).

Random placement of quadrats was achieved by sampling five quadrats per transect along three transects on each sampling date (Resh 1977). The locations of transects and quadrats were selected from a random-number table, and to avoid resampling disturbed areas, no transects were taken within a band 2 m upstream and downstream of a sample site for at least 3 mo. Pool areas, while not avoided, were so rare at the sampling site that virtually all quadrats fell in riffles.

Samples were outlined by a 31.5 × 31.5 cm wire frame secured upstream of a 230-μm mesh drift net. Individual rocks were removed from the sample area, and organisms of interest were collected by hand. If a sample area contained rocks too large to be moved, the net was positioned downstream of the rock, and its surface was washed in front of the net. When suitable placement of the net was impossible or when a significant portion of the area was covered with rocks that protruded above the water surface, the quadrat was moved upstream until the obstacle was cleared. After each rock had been examined, it was washed in front of the net to dislodge any remaining organisms.

Sampling continued until all rocks with diameters >≈5 mm had been washed. The contents of the drift net were then emptied into a white enamel pan, from which any remaining organisms were removed. Specimens were preserved in 3–4% formalin.

Instar and biomass determinations

Larval head capsule widths were measured dorsally across the eyes with an ocular micrometer at 25× or 50×. Instars were identifiable by discrete size-classes with nonoverlapping ranges. The number of larval instars agreed with those reported in the literature: five for the caddisfly species (Anderson and Bourne 1974, Wiggins 1977) and four for the Blephariceridae (Hogue 1973).

Glossosomatid (*Glossosoma* and *Agapetus*) specimens were classified as prepupae if they had removed the ventral strap of their cases, whether or not a cocoon had been spun. Last-instar *Neophylax* close their cases in February–March and aestivate until the following August–September, when pupation occurs (Wiggins 1977). Aestivating *Neophylax* larvae were distinguished from active larvae on the basis of the closed case.

Individual masses were determined for each instar, prepupa, and pupa of each species. Animals were selected from the preserved samples, dried to constant mass at 65°C, and weighed to the nearest microgram on a Cahn Model 21 electrobalance; the mass of the ash was determined after combustion (500°–510°C). Ash-free dry mass (AFDM) was used in all calculations. Standing stocks were calculated for each sample.

A population's voltinism and its cohort production interval (CPI) have a major effect on estimation of its secondary production (Benke 1979, Waters 1979). Knowledge of these two life history factors is especially crucial for use of the size-frequency method (Benke 1979). Summary life cycles were constructed from instar frequency and the mean individual mass per sample.

Cohort production intervals (CPI), required for the size-frequency production calculations, should reflect the average period required for growth, from hatching to the attainment of maximum larval size. Unfortu-

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TABLE 1. Calculation of cohort production intervals (CPI) for six species of stream grazers in Lower Slope Fork, North Carolina, 1979–1980. The CPI has been defined as the period from the sample containing a peak percentage of first instars to the sample containing a peak percentage of prepupae.

Species	Peak abundance (by %) of:				CPI (d)
	First instars	Final instars	Prepupae	Pupae	
Agapetus sp.	1 May*	2–14 June	14 June	2 July	44
Glossosoma nigrior					
Winter	1 December	15 April	1 May	1–19 May	152
Summer	2 June	13 October–1 November	12 December	2 February	192
Goera fuscula	September†	15 March	15 April	2 June	215
Neophylax consimilis	1 December	2 February	15 March‡	15 September	135
Blepharicera williamsae	1 December	15 April	···§	19 May	170
Blepharicera spp.	21 February*	19 May	···§	2 June	102

* First instars not collected quantitatively; sampling date before first collection of second instars used.
† Data from Vannote and Sweeney (1980).
‡ Aestivating V^{th} instars.
§ Prepupae not distinguished from final instars.

nately, given a complex life history, a precise and objective measure of CPI is difficult to obtain. One measure is the interval from the first sample containing first instars to the first sample containing prepupae or pupae. Objections to this measure are: first, that it measures something closer to minimum CPI than average CPI, and second, that it is not practical for species with broadly overlapping cohorts.

The method used here was to identify the samples in which each instar had its peak density. The progression of the peak in time from first to final instar outlines the development of a cohort, and CPIs were calculated from the sample containing peak percentage of first instars to the sample with the peak percentage of prepupae (Table 1). The *Neophylax* CPI, biomass, and production estimates were terminated on 15 March, when a majority of larvae had closed their cases. Only 33 *Goera* were collected in the quantitative benthic samples; consequently, determination of its life history was tenuous. Available data were compatible with the phenology for *Goera calcarata* in Pennsylvania (Vannote and Sweeney 1980), and their September date was used as the period of recruitment.

Production

Detailed knowledge of life histories is essential for the accurate and efficient estimation of secondary production (Waters 1979). Cohort production methodology requires either summing losses between successive samples (the removal summation method) or summing growth over sampling intervals (increment summation and instantaneous-growth methods). These methods assume that a cohort can be unambiguously distinguished in field populations, and that recruitment of first instars is confined to the first sampling interval.

The size-frequency (S-F) method (Waters and Hokestrom 1980) divides a population into size-classes rather than sequential sampling periods, summing losses between size-classes rather than between sampling dates, and does not require synchronous, recognizable cohorts. Comparisons of various methods with simulated and field data (Waters 1977) indicate that the S-F method is comparable to the cohort methods. Krueger and Martin (1980) recently provided an algorithm for calculating confidence intervals for the S-F method, enhancing its usefulness.

One crucial shortcoming of the S-F method is that it does not permit temporal assessment of production. Scraper populations in Shope Fork did not yield interpretable survivorship functions or Allen curves. Field cohorts could not be clearly identified for several of the species. These factors ruled out most of the cohort methods of estimating production. We used the S-F method, as implemented by Krueger and Martin (1980), to provide a baseline estimate that is relatively independent of cohort definition.

To provide seasonal estimates of production, we calculated instantaneous-growth estimates in two stages. First, we fit an exponential growth curve to the mean individual mass, by sample. Production was estimated using the annual form of the instantaneous-growth (IGR) method:

$$P = G \cdot \bar{B},$$

in which G is the annual growth coefficient, and $\bar{B}$ is the mean annual biomass (Ricker 1946). These estimates of P were compared with the S-F estimates to check the overall reasonableness of the growth curves. Second, production over each interval between samples was calculated as:

$$P_i = \frac{\ln(M_i/M_{i-1})}{\Delta t} \cdot (B_{i-1} \cdot B_i)^{1/2},$$

with M_i the mean individual mass, B_i the mean biomass on the ith sampling date, and Δt the time interval (in days) between the $i - 1$ and the ith sampling date (Cummins 1975). Production over the last interval (i.e., from the last sample for which a species was present

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TABLE 2. Growth and mortality curves for stream grazers in Lower Shope Fork. Exponential curves were fit to the data with linearized least squares estimation. The growth curves were fit both to calendar time (number of days elapsed since recruitment) and physiological time (summed degree days, measured above 0°C). Degree days were multiplied by 10^{-1} before equations were calculated. Goodness of fit for the growth and mortality curves was calculated by means of F tests.

	Growth curves ($M = a \cdot e^{gt}$)								Mortality curves ($N = a \cdot e^{-mt}$)			
	Calendar days				10^{-1} × Summed degree-days				Calendar days			
Species	a	g	r^2	n	a	g	r^2	n	a	m	r^2	n
Glossosoma nigrior												
Winter	0.025	0.0311	.79**	12	0.026	0.0567	.73**	11	77.621	0.0281	.78**	8
Summer	0.011	0.0279	.94**	14	0.013	0.0242	.92**	14	1468.290	0.0314	.84**	12
Agapetus sp.	0.009	0.0594	.73**	6	0.094	0.0153	.52	6	39.664	0.0529	.73	5
Neophylax consimilis	0.253	0.0420	.92**	9	0.015	0.0779	.94**	9	16.860	0.0089	.38	8
Goera fuscula	0.044	0.0224	.97**	7	0.032	0.0308	.94**	7	1.743	−0.0013	.04	8
Blepharicera williamsae	0.019	0.0240	.85**	8	0.031	0.0487	.77**	7	7.832	−0.0228	.35	8
Blepharicera spp.†	0.029	0.0159	.37	10	0.057	0.0090	.22	10	161.883	0.0144	.51*	8

* $P < .05$, ** $P < .01$.
† Mixture of at least two species (see Introduction).

to the first sample for which it was absent) was calculated using the average individual mass for the largest size-class (M_i) and calculating the biomass over the interval as half the next-to-last value ($\frac{1}{2}B_{i-1}$).

The IGR model can produce negative production estimates if mean individual mass declines over a sampling interval. The species designated "*Blepharicera* spp.," taken together, had an irregular growth curve, and the fit to the exponential model was poor (Table 2). We assigned this group a cohort production/biomass ratio (P/B) of 5 (Waters 1977) and used this value to calculate the growth coefficient. Because negative values for the other species amounted to only 0.2% of the total production, they were averaged in with adjacent intervals.

RESULTS AND DISCUSSION

Life cycles

Monthly instar distributions of the species are presented in Fig. 2. Sample sizes for *Goera* (33 individuals over 21 sample periods) were too small to permit a meaningful interpretation of its life history. Organisms denoted "*Blepharicera* spp." represent a combination of at least two species and a detailed analysis of this group's life history is not possible at this time.

Early instars of all species were undersampled. *Blepharicera* first instars are extremely difficult to collect, due to their small size and strong adherence to smooth substrates. Preliminary sampling indicated that Surber-type sampling collected only 0–15% of the *Blepharicera* larvae obtained by careful handpicking with fine forceps. No first instars were collected with traditional substrate agitation methods. Even with handpicking, *Blepharicera* first instars must have been undersampled.

No *Agapetus* were collected in the 1 May 1980 sample, while the 19 May sample had roughly equal numbers of second to fourth instars (Fig. 2), indicating extremely rapid larval development. Anderson and Bourne (1974) observed equally rapid larval development by *Agapetus bifidus* in an Oregon stream: first instars were found only in April, and even then second instars predominated; the majority of specimens were pupating by July.

First-instar *Neophylax* occurred in November, and by February aestivating final instars with closed cases were found. Most larvae had closed their cases by 15 March, and no active larvae were found after 1 April. Aestivating larvae showed a marked tendency to aggregate in masses of 5–20 or more individuals, typically under large boulders in deep parts of the stream.

Blepharicera williamsae also had a univoltine life cycle. Larval development began in December and was regular and quite rapid. Final instars were present in substantial numbers by February, and by May most individuals were pupating. No method of identifying *Blepharicera* pupae to species is currently known (Hogue 1978, C. L. Hogue, *personal communication*), and we distinguished *B. williamsae* pupae from the other species by their distinctly larger size.

Glossosoma nigrior had a bivoltine pattern. Both recruitment and pupation were prolonged, making it difficult to separate the overlapping cohorts. First instars were present continuously from May through December, but the peak in October–December and the November peak in second instars suggest October–November recruitment of the winter cohort (Fig. 2). The two cohorts could be separated in December, at which time few final instars were present, and the first to fourth instars could be assigned with some certainty to the winter cohort.

A summer cohort began in May, and most individuals reached the final instar in October–November. the instar distributions for each sampling date (Fig. 2) indicate that some first instars recruited in late April matured by June–July and may have been able to initiate a small second generation overlapping broadly with the principal summer cohort. Temperature data

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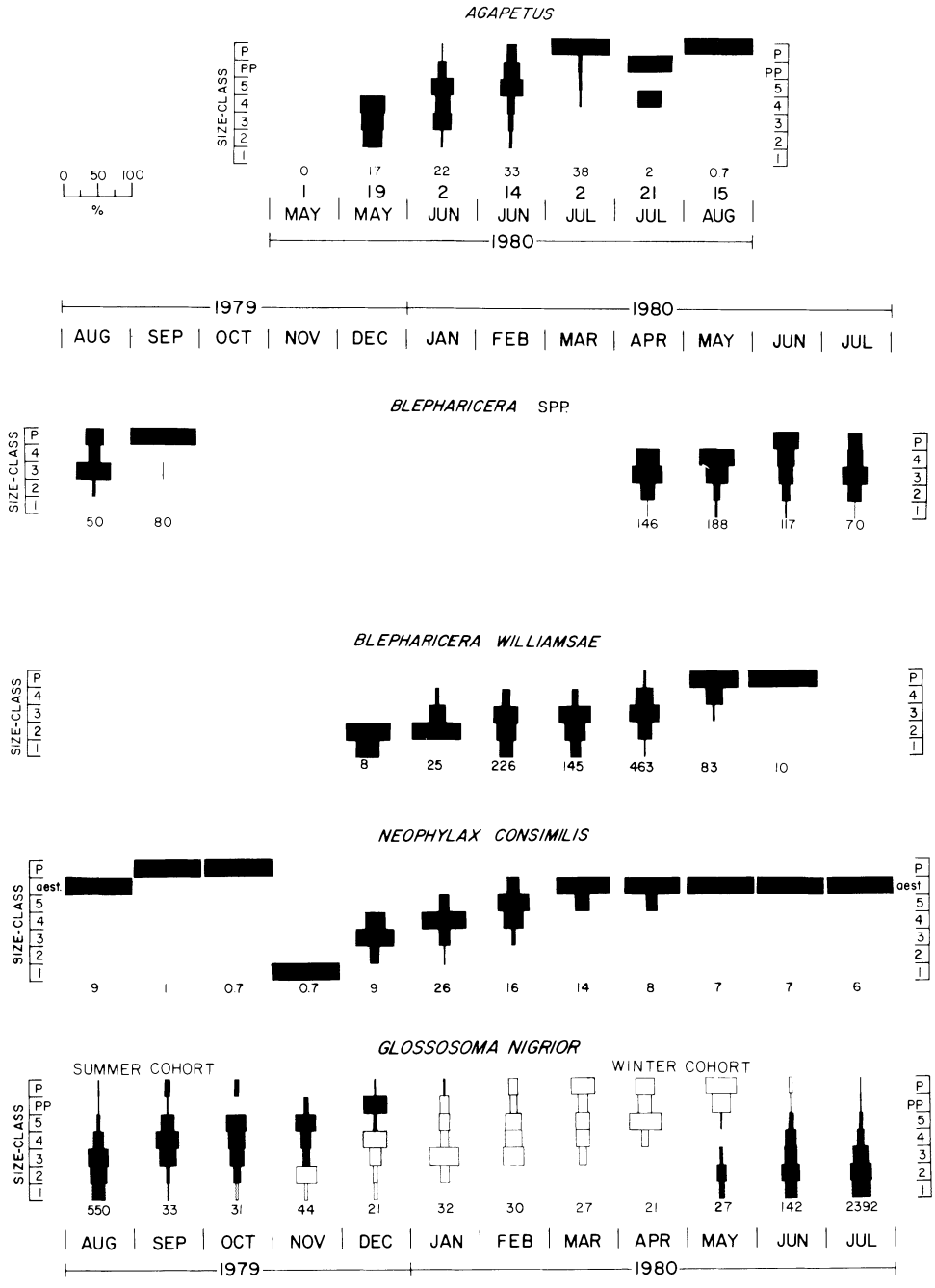

FIG. 2. Monthly instar frequency distributions for five species of scrapers in Lower Shope Fork. Prepupae are designated as PP, pupae as P, and aestivating *Neophylax consimilis* larvae as aest. Monthly values are means of two biweekly samples, except for *Agapetus* sp., for which an expanded time scale has been used. Mean monthly density (no. individuals/m^2) is indicated for each month.

support this interpretation. Degree-days that were measured above 0°C over the period of development for the winter cohort, October through May, total 1539. The degree days summed from 15 April–21 July total 1408, and from 21 July–1 November total 1495, suggesting that at least a partial second summer generation is possible.

Other studies of *Glossosoma* spp. confirm the life history presented here. Tebo and Hassler (1961) sampled *G. nigrior* in Shope Fork and found maximum densities in November and August. Vannote and Sweeney (1980) reported that *G. nigrior* was bivoltine in White Clay Creek, Pennsylvania, with the winter cohort developing from the beginning of October–

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TABLE 3. Mean ash-free dry biomass ($\bar{B}$, mg/m²), annual production (AFDM; P, mg·m^{-2}·yr^{-1}), and production/biomass (P/B) ratios as estimated by the instantaneous-growth and size-frequency methods.

Species	Instantaneous-growth method					Size-frequency method				
	Mean biomass				Cohort	Mean biomass				Cohort
	Annual	Cohort	P	G*	P/B	Annual	Cohort	P	95% CI on P	P/B
Agapetus sp.	0.95	3.67	20.53	21.68	5.59	0.76	3.17	21.55	(14.1–29.0)	6.79
Glossosoma nigrior:										
Winter	11.46	23.41	130.09	11.35	5.56	13.14	25.08	110.74	(77.1–144.4)	4.43
Summer	45.67	75.43	545.09	12.18	7.23	47.68	66.63	501.54	(453.3–549.8)	7.53
Goera fuscula	2.01	2.84	16.43	8.17	5.78	1.46	2.04	9.48	(3.9–15.1)	4.64
Neophylax consimilis	11.49	32.58	176.30	15.34	5.41	9.93	23.18	150.02	(84.0–216.1)	6.47
Blepharicera williamsae	22.76	67.83	199.41	8.76	2.94	26.36	69.19	206.15	(134.3–278.0)	2.98
Blepharicera spp.†	14.60	21.51	107.55	7.36	5†	10.22	17.88	119.12	(87.8–150.4)	6.66

* Instantaneous-growth coefficient (yr^{-1}).
† P and G calculated by assuming the cohort P/B = 5.

March, and the summer cohort from late May through the end of July. Anderson and Bourne (1974) found *G. penitum* in Oregon to be bivoltine, with broadly overlapping cohorts and a life cycle similar to that observed here.

Production

Exponential growth and mortality curves were fit to the data by least squares estimation (Table 2). Undersampling of early instars and high quadrat-to-quadrat variability in density resulted in generally poor fits to the exponential model of mortality. Mortality curves were calculated beginning with the sample with peak first-instar density, thus avoiding negative mortality estimates caused by recruitment. The two significant fits (*Glossosoma* summer and winter cohorts) yielded similar mortality coefficients of 0.028–0.031.

The growth curves were fit to calendar and summed degree-day time (Taylor 1981). The "physiological time" scale based on degree-days did little to improve the fits, and the curves based on chronological time were used to estimate production. The degree-day curves did indicate that the winter species (*Glossosoma* [winter cohort], *Neophylax*, and *B. williamsae*) had higher specific growth rates, relative to temperature, than did the summer species, *Agapetus* and the summer *Glossosoma* cohort (Table 2).

Production estimates and P/B ratios calculated by the S-F and IGR methods are compared in Table 3. All IGR production estimates except that for *Goera* fell within the 95% confidence intervals calculated for the S-F method. Total production (as AFDM) calculated by the annual IGR method was 1195 mg·m^{-2}·yr^{-1}, 6.9% more than the S-F method. The interval IGR method gave a production estimate of 1020 mg·m^{-2}·yr^{-1}, 8.8% less than that calculated using the S-F method. The discrepancy between the two methods of estimating production is in line with that reported in previous comparisons of cohort and size-frequency methods (cf. Wildish and Peer 1981). The P/B ratios calculated by the two methods are in general agreement with simulated and empirical ranges reported for other freshwater invertebrates (Waters 1977). Both methods resulted in an unusually low cohort P/B for *B. williamsae* and in somewhat high values for the summer cohort of *Glossosoma* (Table 3). Considering the irregular mortality curves obtained for most species, the agreement between the IGR and S-F methods is remarkable.

The mean biomass estimates (Table 3) differ between the two methods for two reasons. First, Krueger and Martin's (1980) program uses an annual mean individual mass for each size-class, while we used mass measurements from each sample date in the instantaneous growth calculations. Second, we used the total number of samples in which a species was present to calculate the mean cohort biomass. This period is not necessarily equal to the CPI used in the S-F calculations.

The CPIs used for the two *Glossosoma* cohorts (Table 1) are somewhat uncertain due to the complexity of *Glossosoma*'s life history. They may be compared to values given by Cummins (1975) for *G. nigrior* in Pennsylvania. He reported generation times for two consecutive years of 158 and 197 d for the summer cohort and 269 and 206 d for the winter cohort. His longer winter CPI is probably a result of colder winter and spring stream temperatures in Pennsylvania. Cummins' data indicate that our estimate of the summer cohort's CPI is probably too long, as discussed above. The CPIs for *Neophylax*, *B. williamsae*, and *Agapetus* are clearly defined by the instar frequency distributions (Fig. 2).

Temporal organization

Each species' production (Fig. 3) was concentrated in a short time interval, ranging from <6 wk for *Agapetus* to ≈3 mo for the winter *Glossosoma* cohort. The

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longer period of production for *Blepharicera* spp. is expected, since it is a composite of at least two species. The largest portion of the summer *Glossosoma* cohort's production occurred during July and August when growth was most rapid and biomasses highest. Production at a much lower rate continued through the fall and early winter, as fewer late instars matured.

The species' production peaks occurred at separate points in the year, with remarkably little overlap between species. The winter species, *Glossosoma* (winter cohort), *Neophylax*, and *B. williamsae*, all hatched at about the same time in late fall (Table 1) but exhibited different growth patterns. *Glossosoma* grew steadily from January through March, with a broad and somewhat bimodal production curve (Fig. 3). *Neophylax* and *B. williamsae*, in contrast, had maximum production near the end of their larval period. *Neophylax*'s specific growth rate was nearly twice that of *B. williamsae* (Table 2), with the result that *Neophylax* production peaked in February–March, while most of *B. williamsae*'s production was in late March and April. Production by the spring–summer species was also temporally separated, with *Blepharicera* spp.'s production concentrated in April–May, *Agapetus* in May–June, and the summer *Glossosoma* cohort in July and August.

The significance of the apparent regularity of species' production peaks throughout the year was tested against a random model of phenology developed by Poole and Rathcke (1979). The pattern for all six species, including *Goera*, was significantly more regular than would be expected by chance ($P < .1$).

We calculated temporal overlap among species using proportional similarity as an index of overlap (Whittaker 1975):

$$\alpha_{ij} = \sum^{k} \min(p_{ik}, p_{jk}),$$

where α_{ij} is the overlap between the ith and j^{th} species, and the $p_{.k}$ are the proportions of the species' densities, biomasses, or production which occur over the kth sampling interval. Recent simulation studies (Ricklefs and Lau 1980, Linton et al. 1981) demonstrated that proportional similarity estimates overlap over a wide range (.07–.9), with lower bias and better precision than other commonly used indices. Temporal overlap estimates based on density, biomass, and production are listed in Table 4. The overlaps based on density and biomass were generally similar, with means of .173 and .171, respectively. The mean overlap based on production was .124, significantly lower than the means for density or biomass (paired t test, $P < .05$). The production overlaps are remarkably low, with a maximum of under .3 and with about half <.1. Overlaps based on production should most closely reflect patterns of resource consumption, although most studies of resource utilization are based on density data or, more rarely, biomass.

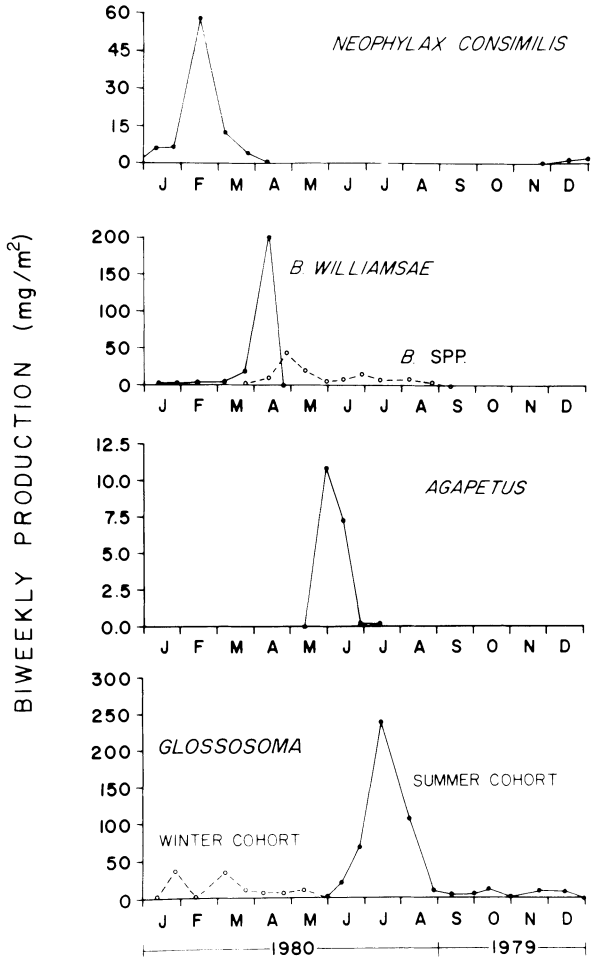

FIG. 3. Secondary production (AFDM) by sample interval for five scraper species. Production over each interval was calculated with the instantaneous growth method. Note the differing vertical scales.

Significance of scraper seasonal production

Combined monthly production by all six scraper species (Fig. 4) was highest in April and June–July. Winter production (January–March) constituted 17.4% of the total annual production. Spring, summer, and fall percentages were 36.2, 38.0, and 8.4, respectively. We anticipated a distinctly different seasonal production pattern, with approximately equal spring and fall maxima, more closely tied to the light levels typical of intermediate-sized streams. McIntire and Colby (1978) summarized the dynamics of a complex trophic model of energy flow in a small woodland stream. They presented seasonal biomass dynamics generated under two light regimes, the actual light schedule in Berry Creek, Oregon, and a hypothetical light schedule sim-

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TABLE 4. Temporal overlaps between species calculated as proportional similarity over 21 biweekly samples.

Species	Overlaps based on densities					
	1	2	3	4	5	6
1. *Glossosoma nigrior* (winter)	...					
2. *Neophylax consimilis*	.668	...				
3. *Blepharicera williamsae*	.442	.379	...			
4. *Blepharicera* spp.	.265	.029	.029	...		
5. *Agapetus* sp.	.030	.000	.000	.318	...	
6. *Glossosoma nigrior* (summer)	.012	.012	.005	.205	.136	...
7. *Goera fuscula*	.373	.310	.103	.187	.000	.140

Species	Overlaps based on biomass					
	1	2	3	4	5	6
1. *Glossosoma nigrior* (winter)	...					
2. *Neophylax consimilis*	.477	...				
3. *Blepharicera williamsae*	.377	.431	...			
4. *Blepharicera* spp.	.410	.039	.039	...		
5. *Agapetus* sp.	.030	.000	.000	.255	...	
6. *Glossosoma nigrior* (summer)	.017	.050	.022	.191	.177	...
7. *Goera fuscula*	.364	.190	.134	.226	.000	.166

Species	Overlaps based on production					
	1	2	3	4	5	6
1. *Glossosoma nigrior* (winter)	...					
2. *Neophylax consimilis*	.299	...				
3. *Blepharicera williamsae*	.192	.116	...			
4. *Blepharicera* spp.	.281	.015	.118	...		
5. *Agapetus* sp.	.000	.000	.000	.146	...	
6. *Glossosoma nigrior* (summer)	.005	.032	.013	.261	.065	...
7. *Goera fuscula*	.288	.245	.259	.172	.000	.092

ulating a wider, less shaded stream. In the Berry Creek simulation grazer biomass peaked in February–early April, declined rapidly toward the end of April, and remained low until September, when it began to rise again to a fall peak. Simulation of a more open stream was similar, except that grazer biomass began to increase in August, and the fall biomass was higher than that over the spring months.

The summer production maximum in Shope Fork was presumably due to the lack of canopy or riparian shading. The extremely low autumn production is surprising in view of the moderate fall temperatures in the region (Fig. 1) combined with the opening of the canopy, even in shaded sites, at leaf fall. However, the temporal production pattern of scrapers in Shope Fork does correlate well with the reported seasonal dynamics of periphyton production. Sumner and Fisher (1979) measured periphyton production in Fort River, Massachusetts. Winter gross primary production (GPP) was low (0.3%) due to low temperatures and ice cover. Spring and summer proportions were roughly equal (48 and 43%), while fall production was only 7% of the annual total. Marker (1976) reported maximum GPP in an intermediate-sized hardwater stream in March–May, with October–December levels much lower. Net primary production in a shallow riffle of the Columbia River was high in March–September and declined in October and November, reaching its lowest levels in the winter months (Cushing 1967).

Minshall et al. (1983) summarized measurements of GPP in first- to seventh-order streams in Pennsylvania, Michigan, Idaho, and Oregon. Of the streams they studied, the third-order Pennsylvania site most closely resembled our study site, based on characteristics such as drainage basin area, gradient, width, and discharge. GPP there was distributed 19% in winter, 27% in spring, 39% in summer, and 15% in fall. Hornick et al. (1981) provided data on net primary production in Guys Run, Virginia, a stream quite similar to our Shope Fork study site except for its spring to early-fall canopy cover. Fall production in Guys Run was actually lower than winter, despite the authors' assumption that production was zero in January and February because of ice cover.

Hall (1972) presented data on both light levels and GPP in a piedmont stream. He found that GPP followed closely the pattern of light availability, with a strong maximum in the early spring, before canopy development. Light levels were much lower in October than in February, despite equivalence of sun angle, due to the prolonged presence of canopy shading. Spring GPP was 12× higher than the autumn maximum, which occurred in October.

In view of the available data on periphyton produc-

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tion dynamics, it would seem that the scrapers in Shope Fork are tracking the general availability of periphyton production in moderate-sized mountain streams reasonably well. The absence of substantial production by these six species from September to December (Fig. 4) is most likely a response to a seasonal minimum in periphyton availability typical of the streams to which these species are adapted, if not necessarily a feature of the unusual conditions present at the study site. The reasons for low periphyton production in moderate-sized streams in the fall are not entirely clear. Temperatures are warmer than in early spring. Depending on the times of canopy development and leaf fall, light levels may be higher in the spring, explaining the difference in seasonal production (Hall 1972). Another possible factor in low fall periphyton production is a seasonal difference in nutrient levels. Monthly nutrient data for Shope Fork indicate no significant seasonal trends except for nitrogen (both NO_3-N and NH_4-N), which is significantly lower from September through December than during the rest of the year (W. Swank, *personal communication*).

The relatively high proportion of scraper production that occurred in the winter may be attributed to two causes. First, unlike most of the streams in the studies reviewed above, Shope Fork is virtually free of ice cover in most years. Second, invertebrates that develop at low temperature lose less energy to respiration, both directly because of the low temperatures and indirectly because they tend to attain larger sizes than the same species growing at warmer temperatures (Vannote 1978, Vannote and Sweeney 1980). Larger individuals have lower mass-specific respiration rates and are able to allocate more of their assimilated energy to tissue elaboration (Vannote and Sweeney 1980). *Glossosoma* prepupae reached a maximum mean AFDM of 3.67 mg/individual in March, as compared to 2.10 mg/individual in August. Sweeney and Vannote (1978) reported similar data for *G. nigrior* in Pennsylvania. Thus, winter scraper populations may make more efficient use of the available primary production.

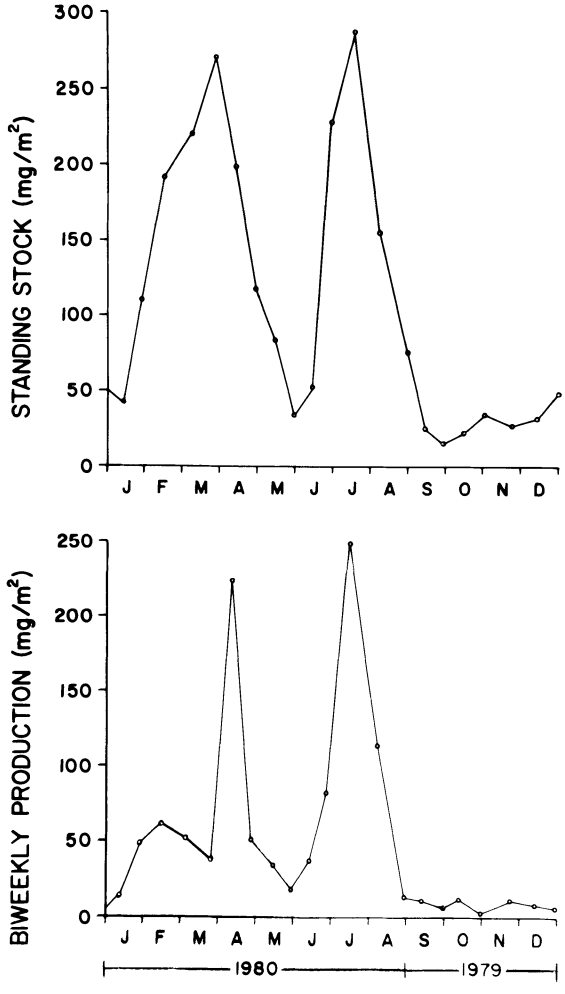

FIG. 4. Combined standing stock and production (AFDM) of six species of scrapers in Lower Shope Fork, 1979–1980. Production of *Goera fuscula* has been added to production of the five species presented in Fig. 3.

CONCLUSIONS

Our results support Vannote's (1978) hypothesis that the species of functional groups within particular stream communities are organized seasonally so as to minimize the periods in which similar resources are being used by two or more species. The assumption that the underlying cause of this temporal organization is the avoidance of interspecific competition has not been tested. Similar patterns have been observed in another group of periphyton-grazing insects and among six species of *Ephemerella* mayflies in White Clay Creek, Pennsylvania (Vannote and Sweeney 1980, Sweeney and Vannote 1981). Temperature is presumably the underlying factor that governs the seasonal organization of scraper species in Shope Fork, although we do not have the temperature-specific bioenergetic data required to verify the role of temperature.

The pattern of temporal segregation observed requires close timing of the life cycles of the species involved. Three of the six species possess unusual life history features. The larval development period of *Agapetus* was ≈1½ mo. The similar life cycles of *A. bifidus*, *A. occidentis*, and *A. illini* have been attributed to a 9–10 mo egg diapause (Anderson 1976, Wiggins 1977). *Neophylax* larvae in Shope Fork closed their cases in late March and began an ≈6-mo larval aestivation (Fig. 2), a typical life history for *Neophylax* species (Mackay and Wiggins 1979). *Goera* was too uncommon to permit a quantitative assessment of its life history. Our data do indicate that *Goera* fifth instars maintained approximately constant individual mass from November through February, with rapid

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individual growth from 2–3 to 7–8 mg AFDM in late February and March. These data are compatible with Vannote and Sweeney's (1980) observation of a mid-winter larval growth cessation in *Goera calcarata* in Pennsylvania.

Glossosoma did not exhibit a diapause period, but the winter cohort's abundances and production were considerably lower than those of the summer cohort (Fig. 3). This asymmetry between the cohorts may be an important factor in *Glossosoma*'s dominance of the scrapers' production, since it shifts resource utilization away from the winter months when temporal overlap with other species would be greater (Table 4). The lack of canopy cover at the study site presumably results in much higher late-spring and summer periphyton growth than would occur in more typical fourth-order streams in the eastern deciduous biome. Apparently the univoltine species lack the life history flexibility required to take advantage of a locally available resource at an unusual time of year. *Glossosoma* can, in contrast, utilize the summer periphyton crop without a major alteration of its life history, since its summer cohort is adapted to summer temperature regimes. Despite lower growth rates (as measured on a degree-day scale, Table 2), the summer *Glossosoma* cohort has 4.5× the production of the winter cohort. *Glossosoma*'s bivoltinism, in combination with long periods of diapause in the other species, resulted in a sequence of production by various species throughout the seasons in which the food resource is likely to be abundant.

Grant and Mackay (1969) also found temporal partitioning to be the predominant mode of ecological segregation among groups of systematically related stream insects and concluded that energy flow through congeneric species should be temporally staggered throughout the year. Our data from Shope Fork extend this concept to a functionally similar but systematically diverse group of species. Schoener (1974) surveyed a wide range of studies of ecological segregation and found that 44% of invertebrates showed seasonal segregation, compared to only 13% of vertebrates studied. The possession of life cycles less than a year in duration may be an advantage to organisms in environments with predictable seasonal fluctuations in resource type and availability.

The data indicated that under the optimum conditions present at the study site, i.e., a riffle habitat with no canopy shading, grazers can play an important role in energy flow in streams of intermediate size. The total secondary production (AFDM) of 1.12 $g \cdot m^{-2} \cdot yr^{-1}$ is of the same magnitude as production estimates for filter feeders in similar streams nearby (Benke and Wallace 1980, Ross and Wallace, *in press*), and the values for individual species (Table 3) span the mode of production values reviewed by Waters (1977). The production of exuviae and secretions (e.g., silk), which we did not measure, could increase the total production by as much as 30% (Castro 1975). Similar species of scrapers studied by Neves (1979) produced dry mass of ≈ 0.5 $g \cdot m^{-2} \cdot yr^{-1}$, while Cummins (1975) reported production (AFDM) by *Glossosoma nigrior* alone in a Pennsylvania stream of 0.9 – 1.3 $g \cdot m^{-2} \cdot yr^{-1}$. (AFDM data were expressed in kilocalories by Cummins [1975]; 1 g = 5.7 kcal.)

The complex nature of epilithic accumulations (Perkins and Kaplan 1978) makes it difficult to distinguish the trophic basis of scraper production. We examined qualitatively gut contents of the scraper species in Shope Fork. Intact diatom frustules and amorphous detritus clumps occupy roughly equal relative areas, but the abundance of diatoms varies widely among specimens and species. Of particular interest is the observation that what appear to be amorphous clumps when viewed under normal lighting and low magnifications can be seen to consist of masses of small (on the order of 20 μm long) diatom frustules when viewed at 400× with phase microscopy. Given the difficulty of distinguishing gut materials, we assume that a substantial portion of the energy basis of scraper production in Shope Fork is derived from autochthonous sources, either directly or in the form of autochthonous detritus (Naiman 1976, Perkins and Kaplan 1978). Assuming gross growth efficiencies of $\approx 15\%$ on algal material (Benke and Wallace 1980), primary production of ≈ 8 $g \cdot m^{-2} \cdot yr^{-1}$ would be required as input. Algal material can also form a considerable portion of the diet of organisms classified as collector-gatherers (cf. Kondratieff and Voshell 1981). Ross and Wallace (1981) analyzed gut contents of *Brachycentrus spinae* in Shope Fork. Algal material constituted nearly 20% of the gut contents of this species, which is usually categorized as a collector.

Hornick et al. (1981) estimated GPP (as Carbon) in a stream similar to Shope Fork as 6.54 $g \cdot m^{-2} \cdot yr^{-1}$ (or as AFDM, ≈ 14 $g \cdot m^{-2} \cdot yr^{-1}$). They compared GPP in shaded and unshaded stream sites and found that at fully canopy, production was $\approx 3\times$ higher at the unshaded site. McIntire and Colby's (1978) stream simulation model increased annual GPP twofold when an open-canopied light environment was simulated. A reasonable upper estimate of annual GPP in Shope Fork is therefore on the order of 30–35 g/m^2, of which scraper ingestion would constitute $\approx 25\%$. Sloughing losses and ingestion of periphyton by species which function partially as collector-scrapers (e.g., *Ephemerella* and heptageniid mayflies) would result in a closer correspondence between scraper production and resource availability. Filter feeders in similar southern Appalachian streams, by contrast, ingest only a tiny proportion of the available particulate organic matter in transport (Benke and Wallace 1980, Georgian and Wallace 1981, Ross and Wallace, *in press*).

Stream periphyton provides a mechanism by which dissolved nutrients that otherwise would be lost to downstream reaches are instead removed from the

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water column and made available to benthic consumers (Cushing 1967, Elwood and Nelson 1972). Grazers may actually enhance periphyton production (Kehde and Wihlm 1972, Cooper 1973) or food quality (Hunter 1980), while the retention of nutrients in lotic ecosystems can be decreased by the activities of shredders (McDiffett and Jordan 1978, Wallace et al. 1982). In nutrient-poor streams such as Shope Fork, which are dependent on allochthonous nutrient inputs, the periphyton-grazer pathway may contribute more to nutrient spiralling (Webster and Patten 1979) than an analysis of energy flow alone would indicate.

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